

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/387810844>

A style transfer-based augmentation approach for detecting military camouflaged object

Article in *Journal of Military Science and Technology* · December 2024

DOI: 10.54939/1859-1043.j.mst.CSCE8.2024.44-54

CITATIONS

3

READS

84

2 authors:



Hang Thi Thu Truong

AMST

7 PUBLICATIONS 13 CITATIONS

[SEE PROFILE](#)



Kien Tran

Information Technology Institue

26 PUBLICATIONS 194 CITATIONS

[SEE PROFILE](#)

A style transfer-based augmentation approach for detecting military camouflaged object

Truong Thi Thu Hang, Tran Trung Kien*

Institute of Information Technology, Academy of Military Science and Technology, 17 Hoang Sam, Cau Giay, Hanoi, Vietnam.

*Corresponding author: t2kien@gmail.com

Received 11 Oct. 2024; Revised 07 Dec. 2024; Accepted 12 Dec. 2024; Published 30 Dec. 2024.

DOI: <https://doi.org/10.54939/1859-1043.j.mst.CSCE8.2024.44-54>

ABSTRACT

Detecting camouflaged objects in military environments is particularly challenging due to the deliberate blending of targets with their surroundings. Despite advancements in deep learning, the limited availability of training data remains a significant obstacle. To address this issue, this paper proposes a novel data augmentation approach that combines multiple style transfer models to transform the style of training images into diverse reference styles. This enriches the training data by simulating various environmental textures and patterns. Structural Similarity (SSIM) is used as an evaluation metric to select style-transferred images that preserve the best structural similarity, enabling detection models to more effectively learn the distinguishing features of camouflaged objects. Extensive experiments with different style transfer methods and SSIM thresholds show that our augmentation approach significantly enhances the accuracy of state-of-the-art detection algorithms. This approach has the potential to improve object detection in military operations, increasing the reliability and precision of automated surveillance systems.

Keywords: Style transfer; Military camouflaged object detection; Data augmentation.

1. INTRODUCTION

Camouflage is a survival strategy observed extensively in nature, where animals blend with their environment to evade predators or ambush prey. The effectiveness of natural camouflage relies on its ability to obscure the object from detection by mimicking patterns, colors, and textures of the surrounding environment. Inspired by nature, military forces have adopted camouflage techniques to conceal personnel, vehicles, and equipment in diverse environments. Military camouflage aims to reduce the visual and infrared signatures of objects, making detection by adversaries and automated systems extremely difficult.



Figure 1. Some samples from the natural camouflage dataset CAMO (left) [9] and the military camouflage dataset MHCD2022 (right) [12].

Animals, unlike camouflaged military objects, are not hidden by external structures,

leaving their physical features exposed and allowing object detection models to identify them with relative accuracy. In contrast, human-designed military camouflage is quite different. Soldiers, tanks, and vehicles are disguised using color schemes or covered with materials like tarps, grass, or leaves, which obscure important edge and texture features. This makes detection more difficult for models trained on military-specific datasets, as shown in figure 1.

In recent years, the task of detecting camouflaged military objects has drawn significant attention within the military field [3, 10, 12, 14]. For instance, electronic reconnaissance systems deployed on UGVs or UAVs can leverage these solutions to identify the locations of enemy camouflaged objects across various strategic scenarios. However, despite advances in deep learning-based object detection, identifying camouflaged targets remains a challenging problem. Camouflaged objects hide in environments with high similarity and invisibility, so the subtle differences between camouflaged objects and their backgrounds often lead to high false-negative rates, particularly when detection systems rely on limited training data. In military contexts, data collection is constrained by security concerns, varying terrain conditions, and the scarcity of real-world samples across different seasons, terrains, and weather conditions.

In this paper, we introduce a data augmentation approach via the style transfer method to address the problem of limited data and imbalance classes for camouflaged object detection in the military. To summarize, our primary contributions can be outlined in three key points as follows:

- Proposing a style transfer-based augmentation method in military camouflage object detection.
- Combining multiple style transfer networks using SSIM histogram visualization and an SSIM-based method for evaluating and selectively filtering out low-quality images from the style-transferred dataset.
- Conducting experiments to demonstrate that augmenting datasets with style-transferred images significantly enhances the camouflage object detection performance of deep learning models.

The remainder of the paper is organized as follows: Section 2 reviews related work on military camouflaged object detection and style transfer-based augmentation. Section 3 explains the methodology, while section 4 presents the experimental and section 5 shows results and discussion. Lastly, section 6 concludes the paper and suggests potential directions for future research.

2. RELATED WORK

2.1. Military camouflaged object detection

Military camouflaged object detection (MCOD) is a complex task aimed at identifying and recognizing objects, such as personnel and equipment, that are hidden using camouflage techniques in military environments. Recent advancements in deep learning and artificial intelligence have led to the development of specialized algorithms and datasets designed for military applications. Peng et al. proposed a network with an attention mechanism for detecting camouflaged soldiers, using pyramidal feature reduction [14]. Combining military and non-military datasets was a data augmentation

method to enhance camouflaged object detection in the military using deep learning [8]. Liu et al. Introduced an advanced neural network architecture for camouflage object detection and constructed a high-level camouflage benchmark MHCD2022 dataset in the military [12]. Other methods employed synthetic data [3] or based on image style transfer to augment training datasets and improve detection accuracy across different military settings. Despite these advancements, challenges remain in terms of result effectiveness, object class diversity, and detecting highly camouflaged objects.

2.2. Style transfer-based augmentation

Image style transfer is a very prominent research direction in the computer vision and image processing communities, and extensive research has been conducted in this field and achieved significant advancements. It involves applying the style of a reference image to re-render a content image, allowing the resulting image to maintain the original content while adopting the style of the reference image. As a result, style transfer can serve as a data augmentation technique, and many prior studies have employed it to address data scarcity in various tasks [9, 15, 20]. Huang et al. [6] proposed a style transfer-based augmentation framework to improve the performance of deep learning models across different ultrasound sources with limited data. In cross-domain few-shot learning, Rao et al. [15] exploited style transfer methods to enhance the domain generalization ability. While numerous studies have explored data augmentation using style transfer in deep learning applications, research in the military domain, particularly on military camouflage, remains limited. Lee et al. applied FastPhotoStyle [10], a pre-trained style transfer model, to synthetic data for military vehicle detection. Zhou et al. [21] employed style transfer in domain adversarial learning to augment data for military vehicle detection.

2.3. SSIM-based data augmentation

Structural Similarity Index Measure (SSIM) [17] is a metric used in deep learning and image processing to evaluate the similarity between two images. It plays an important role in data augmentation by ensuring that augmented images maintain perceptual similarity to the original data. Unlike pixel-based metrics, SSIM focuses on comparing structural features, luminance, and contrast, preserving key visual elements such as textures and edges. This helps generate high-fidelity synthetic data, which introduces useful variations without compromising essential details. SSIM-based augmentations help with imbalanced image datasets [5], to selectively filter out the irrelevant data [16], enhance model generalization and robustness, improving performance in tasks like image classification [18], object detection [2], using the average of SSIM as a threshold value to retain more profile information [19]. It is especially valuable in domains like medical imaging [13], where maintaining the structural integrity of augmented images is critical for accuracy.

3. METHODOLOGY

3.1. Overview of the proposed method

Figure 2 illustrates the overall multi-phase process of our proposed method for military camouflage object detection. The first two phases focus on creating a stylized camouflage dataset by combining style transfer methods and refining the stylized images dataset using SSIM threshold. The third phase involves training and validating deep learning models using augmented datasets that integrate both synthetic and real-world data. The details of

each phase will be introduced in the following sections.

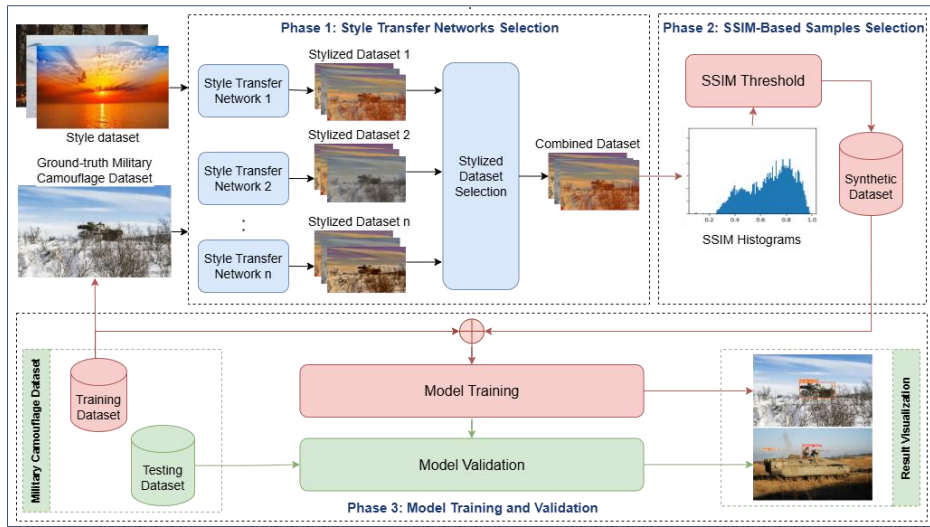


Figure 2. Scheme of the proposed algorithm.

3.2. Style transfer network selection

We use various style transfer networks to generate synthetic camouflage images, blending real camouflage data with seasonal and weather-based styles, such as sunshine, rain, and snow, to create more diverse training examples. The content dataset consists of real camouflage training images, while the style set includes seasonal and weather imagery. For each style transfer model, the generated images form a stylized synthetic dataset.

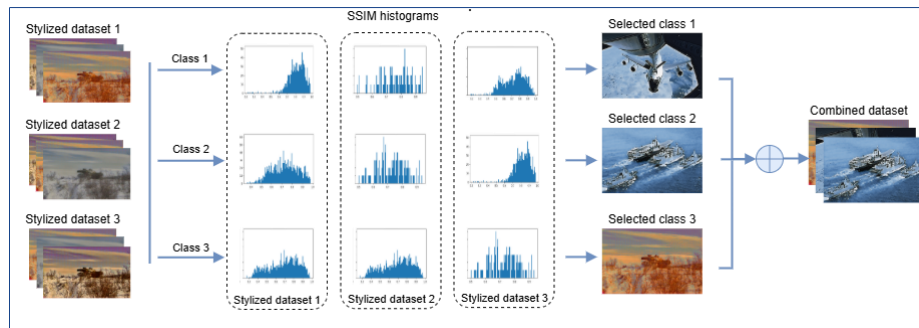


Figure 3. Stylized dataset selection.

We consolidate multiple stylized datasets into a single dataset using the object class selection method. The process is outlined in figure 3 as follows:

Step 1: For each stylized dataset, we compute average SSIM metrics for each object class by comparing the stylized object classes with their counterparts in the original dataset.

Step 2: We generate the SSIM histogram for each object class and examine the distribution to identify the best dataset for that object class.

Step 3: By merging the selected datasets for each object class, we obtain a combined dataset that incorporates various style transfer methods.

3.3. SSIM-based sample selection

Using the combined dataset from phase 1, we calculate the SSIM histogram to analyze

the distribution and select an appropriate SSIM threshold. This threshold is used to filter out images with low SSIM values, as these images have weak structural integrity, potentially reducing the effectiveness of the detection model. A higher SSIM threshold results in fewer images in the final dataset, while a lower threshold increases the number of images. A good starting point for selecting the threshold is the dataset's median SSIM value, which can then be adjusted up or down to find the optimal value for the problem [19, 24]. This process ensures the selection of the best-quality images from the combined dataset.

3.4. Model training and validating

This phase involves creating a high-quality dataset by integrating stylized camouflage data generated in Phase 2 with the original ground-truth training data. To further enhance the dataset's diversity, traditional augmentation techniques, such as rotation, scaling, and flipping, are applied. The testing dataset is strictly composed of the ground-truth camouflage set, which is reserved specifically for the validation stages, preserving its integrity for accurate performance evaluation. This evaluation highlights the improvements brought by synthetic data augmentation and confirms the reliability of the generated dataset for object detection tasks in military applications.

4. EXPERIMENTS

4.1. Dataset

All experiments in this study are performed using the benchmark MHCD2022 dataset introduced by Maozhen Liu et al. in 2023, which represents the first comprehensive initiative dedicated to the detection of military high-level camouflaged objects.

In this experiment, we selected a set of 10 style images representing various weather conditions and seasons, including summer sunshine, winter rain, and snow. To address the class imbalance in the MHCD2022 dataset, we extracted 366 images from the original camouflage training set, featuring objects from four classes: tank, military vehicle, warship, and aeroplane. This collection will be referred to as the "original dataset" from this point onward. Each style transfer model generates a total of 3,660 images. The distribution of objects in the style transferred dataset across specific classes is illustrated in figure 4.

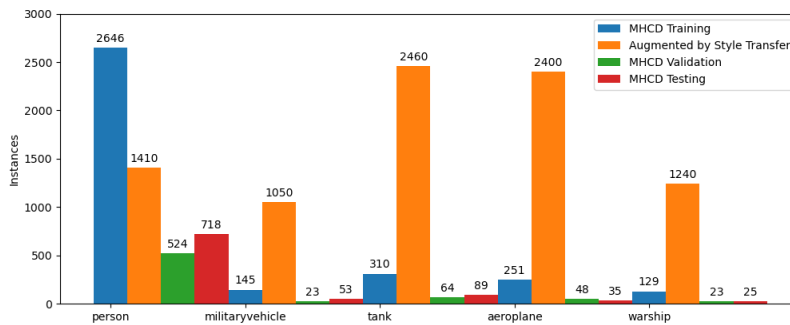


Figure 4. Statistics on the number of instances in each class of the ground-truth military camouflage dataset MHCD2022, along with the synthetic dataset by style transfer.

4.2. Baselines and implementation details

4.2.1. Style transfer

To assess the impact of various style transfer methods on military camouflage

augmentation, we selected several representative style transfer networks. Neural style transfer techniques, which transfer artistic styles from style images to content images, are a key focus. These methods utilize neural flow, a subclass of deep generative models that learn the exact likelihood of high-dimensional data through a series of reversible transformations. Notable examples include ArtFlow [1], which integrates the AdaIN [7] or WCT [11] transfer modules, and the more recent HierarchyFlow [4]. Additionally, we incorporate CycleGAN [22], a widely used GAN model for unpaired image-to-image translation, which leverages cycle consistency loss to preserve content while transferring styles between domains. By default, all images are resized to 512x512 and evaluation is performed in 640x640. We train the style transfer models using the experimental configurations specified by their original authors, with some modifications. HierarchyFlow is trained with the loss weight λ of style loss is set to 0.1 and k in aligned-style loss is set to 1.0. We train ArtFlow in conjunction with the transfer module AdaIN and WCT and content and style loss weight are set to 0.9 and 0.1, respectively.

4.2.2. Object detection

To validate the effectiveness of the entire pipeline, we use YOLOv8 with default hyperparameter settings supported by Ultralytics [23] with an input image size of 640x640 and pre-trained on COCO dataset.

4.3. Evaluation metrics

We use a widely-used metric Mean Average Precision (mAP) to measure the accuracy of YOLO object detection methods, encompassing two types of mAP: mAP@.50, mAP@[.5:.95]. These metrics are employed to assess each class and average of all five classes of the dataset.

5. RESULTS AND DISCUSSION

5.1. Evaluation of multiple style transfer selection

Figure 5 presents SSIM histograms for the Tank, Military Vehicle, Aeroplane and Warship classes across four different style transfer methods: HierarchyFlow, CycleGAN, ArtFlow + AdaIN, and ArtFlow + WCT. Each histogram visualizes the distribution of SSIM values within the stylized datasets, providing insight into the structural similarity between the original and stylized images.

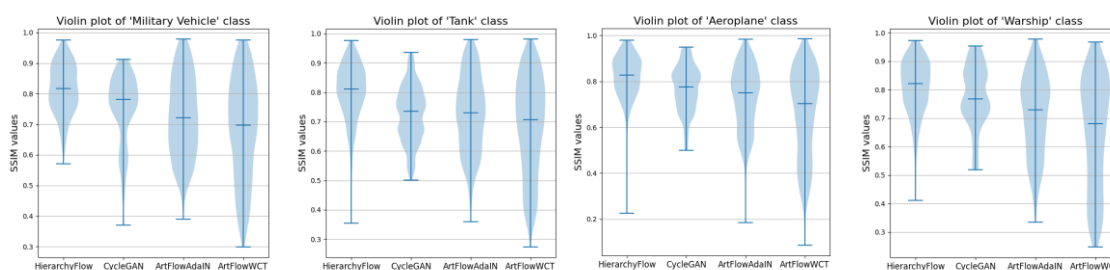


Figure 5. SSIM histograms for each class across four different style transfer methods.

- Military vehicle class: HierarchyFlow maintains higher structural similarity with a concentration around 0.7 to 0.95, while CycleGAN with most values concentrated around 0.7 to 0.9. ArtFlows have the broadest ranges, including many lower values,

indicating a less reliable preservation of structure.

- Tank class: The SSIM values of HierarchyFlow are skewed toward the higher range, with most values concentrated around 0.8 to 0.9. This indicates that HierarchyFlow consistently preserves structural similarity, which aligns with that its highest average SSIM score. The SSIM values range widely from 0.6 to 0.9, suggesting that CycleGAN produces more variability in structural preservation while ArtFlow with a lower concentration.
- Aeroplane class: HierarchyFlow is concentrated towards the higher end, while CycleGAN is spread out around 0.5 to 0.9. ArtFlows have wider ranges.
- Warship class: HierarchyFlow concentrate around around 0.5 to 0.95. CycleGAN focus around 0.7 to 0.9 while ArtFlow show more evenly spread SSIM distributions across the mid-range 0.5 to 0.9.

Overall, HierarchyFlow appears to offer the best structural similarity preservation for the Tank and Aeroplane classes. In the case of Aeroplane, ArtFlowAdaIN can serve as a suitable alternative to HierarchyFlow. For Military Vehicle and Warship classes, CycleGAN performs well, especially in the Warship class. However, both ArtFlowAdaIN and ArtFlowWCT exhibit greater variability, indicating less consistent SSIM performance across different object types. Thus, based on visual analysis, combining the Tank, Military Vehicle, and Aeroplane classes from HierarchyFlow with the Warship class from CycleGAN will result in a dataset of optimal quality, preserving the highest structural similarity to the original dataset (SSIM histogram is shown in figure 6).

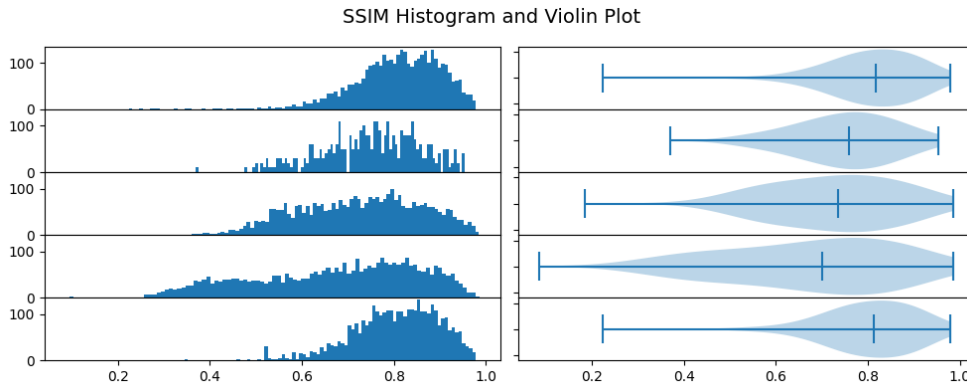


Figure 6. Comparing SSIM histogram of datasets stylized by HierarchyFlow, CycleGan, ArtFlowAdaIn, and ArtFlowWCT and combined dataset, respectively (from top to bottom).

5.2. Performance of the proposed method

After step combining multiple style transfer, we get an optimal structural dataset that combine the Tank, Military Vehicle, and Aeroplane classes from HierarchyFlow with the Warship class from CycleGAN. As shown in figure 6, the median SSIM value of this combined dataset compared to the original dataset is 0.8128. Accordingly, we choose a range of SSIM threshold values including 0.85, 0.8, 0.75, 0.7, and 0.65. Images in the combined dataset with SSIM values below the selected thresholds will be filtered out. This filtering process reduces the combined dataset of 3,660 images to 1,310, 1,994, 2,680,

3,198, 3,421, and 3,542 images, respectively. Using a threshold higher than 0.85 filters out too many images, while a threshold below 0.65 filters too few, resulting in the filtering process ineffective in both cases.

For each selected dataset, we combine it with the original training dataset and train the YOLOv8 object detection model. We refer this final training dataset as MHCD+HF+CycleGAN. The table 1 below presents the results of testing the trained model on the test set, including a comparison with the results of the model trained solely on the original dataset (MHCD).

Table 1. Comparison of mAP implemented by YOLOv8 on the original MHCD dataset, added combined dataset with different threshold SSIM.

Dataset	mAP@.50						mAP@ [.5:.95]
	Person	Military Vehicle	Tank	Aero plane	War ship	All	All
MHCD	0.869	0.432	0.788	0.718	0.710	0.703	0.447
MHCD+HF+CycleGAN	0.837	0.496	0.749	0.846	0.737	0.733	0.492
MHCD+HF+CycleGAN SSIM 0.85	0.848	0.530	0.778	0.840	0.728	0.745	0.493
MHCD+HF+CycleGAN SSIM 0.8	0.842	0.475	0.733	0.701	0.766	0.703	0.453
MHCD+HF+CycleGAN SSIM 0.75	0.823	0.465	0.757	0.925	0.787	0.751	0.496
MHCD+HF+CycleGAN SSIM 0.7	0.834	0.451	0.738	0.834	0.761	0.724	0.465
MHCD+HF+CycleGAN SSIM 0.65	0.834	0.465	0.729	0.825	0.703	0.711	0.462

Based on the results presented in the table 1, several conclusions can be drawn regarding the performance of the object detection model with different datasets:

- The combination of MHCD+HF+CycleGAN SSIM 0.75 (SSIM threshold is 0.75) gives the best overall [mAP@.50](#) score of 0.751, which is an improvement over the original MHCD dataset's score of 0.703. The highest mAP@[.5:.95] score is also achieved with a value of 0.496, indicating that this dataset performs well across a range of IoU thresholds.
- The combination of MHCD+HF+CycleGAN SSIM 0.75 dataset not only achieves the best overall mAP@.50 but also excels in aeroplane (0.925) and warship (0.787) detection, making it a well-balanced dataset for multiple object types.
- The best performance for military vehicle detection comes from the MHCD+HF+CycleGAN SSIM 0.85 set, with a score of 0.530, which is a significant improvement over the original dataset. Although the MHCD+HF+CycleGAN SSIM

0.75 set is not the top performer for this class, it still surpasses the performance of the original MHCD.

Overall, despite some inconsistencies in performance across certain metrics, the results clearly show the effectiveness of our proposed method in enhancing original data for camouflage military object detection (figure 7). Furthermore, they underscore the superior potential of our approach in advanced camouflage scenarios, suggesting strong promise for future integration into individual combat systems.



Figure 7. Some results of camouflaged tank detection trained on distinct datasets: the original MHCD training set (MHCD), added HierarchyFlow-based stylized dataset (MHCD+StylizedHF), added CycleGAN-based stylized dataset (MHCD+StylizedCycleGAN), added combined dataset (MHCD+HF+CycleGAN), and added combined dataset with SSIM threshold 0.75 and ground-truth images with bounding box labels in the first column for a clearer comparison.

6. CONCLUSIONS AND FUTURE WORKS

In this paper, we introduced a novel approach for military camouflage object detection using style transfer techniques, enabling the generation of augmented data across various scenarios from the original training dataset. Additionally, we investigated the impact of different style transfer methods on data augmentation, employing the SSIM metric with various thresholds to validate and select the best augmented images. This approach demonstrated improvements in both the quality and diversity of the camouflage data. Through our experiments, we showcased the effectiveness of style transfer in enhancing military camouflage datasets, significantly boosting detection performance. Consequently, it forms the foundation for advancing highly effective target detection and tracking solutions for electronic reconnaissance systems deployed on UGVs, UAVs for different tasks such as automated patrol, threat identification in the military domain.

Future Work: One limitation of this study is that the results for the 'Tank' and 'Person' classes showed a slight decline, while other classes and overall performance improved. In future work, we plan to address this by conducting a deeper evaluation method using SSIM threshold for each classes. Another limitation is our reliance on existing style transfer techniques, which may not effectively transfer camouflage structures. Therefore,

we aim to develop new techniques or explore hybrid methods that integrate this style transfer framework with generative models for data augmentation. These are promising directions for improving the efficiency and reliability of object detection systems in complex military environments.

Acknowledgment: This research is part of a 2024 scientific research project for young staff at the Academy of Military Science and Technology, titled: "Research on developing AI-powered autonomous navigation and ground target tracking modules deployed on the Warthog UGV".

REFERENCES

- [1]. An, Jie, Siyu Huang, Yibing Song, Dejing Dou, Wei Liu and Jiebo Luo. "ArtFlow: Unbiased image style transfer via reversible neural flows." 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR): 862-871, (2021).
- [2]. Du, Siyu, Cuili Yao, Yuqiu Kong and Yicheng Yang. "BANet: Camouflaged object detection based on boundary guidance and multiple attention mechanisms." 2023 9th Annual International Conference on Network and Information Systems for Computers (ICNISC): 464-469, (2023).
- [3]. Eker, Thijs A., Friso G. Heslinga, Luca Ballan, Richard J. M. den Hollander and Klammer Schutte. "The effect of simulation variety on a deep learning-based military vehicle detector." Security + Defence, (2023).
- [4]. Fan, Weichen, Jinghuan Chen and Ziwei Liu. "Hierarchy flow for high-fidelity image-to-image translation." ArXiv abs/2308.06909, (2023).
- [5]. Guo, Chen, Fabian Benitez-Quiroz, Qianli Feng and Aleix M. Martínez. "Structural similarity: When to use deep generative models on imbalanced image dataset augmentation". (2023).
- [6]. Huang, Bin, Ziyue Xu, Shing-Chow Chan, Zhong Liu, Huiying Wen, Chao Hou, Qicai Huang, Meiqin Jiang, Changfeng Dong, Jie Zeng, Ruhai Zou, Bin Huang, Xin Chen and Shuo Li. "A Style transfer-based augmentation framework for improving segmentation and classification performance across different sources in ultrasound images." International Conference on Medical Image Computing and Computer-Assisted Intervention, (2023).
- [7]. Huang, Xun and Serge J. Belongie. "Arbitrary style transfer in real-time with adaptive instance normalization." 2017 IEEE International Conference on Computer Vision (ICCV): 1510-1519, (2017).
- [8]. Hwang, Kyo-Seong and Jungmok Ma. "Military camouflaged object detection with deep learning using dataset development and combination." The Journal of Defense Modeling and Simulation: Applications, Methodology, Technology (2024).
- [9]. Kanchanadevi, st K. and G. K. Sandhia. "A neural style transfer network for image augmentation of corn leaves using convoluvational neural netwok." 2023 International Conference on Integrated Intelligence and Communication Systems (ICIICS): 1-5, (2023).
- [10]. Lee, Hyeongkeun, Kyungmin Lee, Hunmin Yang and Se-Yoon Oh. "Applying FastPhotoStyle to synthetic data for military vehicle detection." 2020 20th International Conference on Control, Automation and Systems (ICCAS): 137-140, (2020).
- [11]. Li, Yijun, Chen Fang, Jimei Yang, Zhaowen Wang, Xin Lu and Ming-Hsuan Yang. "Universal style transfer via feature transforms." Neural Information Processing Systems (2017).
- [12]. Liu, Mao Zhen and Xiaoguang Di. "Extraordinary MHNet: Military high-level camouflage object detection network and dataset." Neurocomputing 549: 126466, (2023).
- [13]. Maruyama, Sho. "Properties of the SSIM metric in medical image assessment: correspondence between measurements and the spatial frequency spectrum." Physical and Engineering Sciences in Medicine 46: 1131-1141 (2023).

- [14]. Peng, Yiguo, Jianzhong Wang, Zibo Yu, Yu You and Yong Sun. “Camouflage soldier object detection network based on the attention mechanism and pyramidal feature shrinking.” Multimedia Tools and Applications, (2024).
- [15]. Rao, Shuzhen, Jun Huang and Zengming Tang. “Exploiting style transfer-based task augmentation for cross-domain few-shot learning.” ArXiv abs/2301.07927 (2023).
- [16]. Raveendran, Ritin, Aviral Kumar Singh and M RajeshKumar. “Complex wavelet SSIM based image data augmentation.” ArXiv abs/2007.05853 (2020).
- [17]. Wang, Zhou, Alan Conrad Bovik, Hamid R. Sheikh and Eero P. Simoncelli. “Image quality assessment: from error visibility to structural similarity.” IEEE Transactions on Image Processing 13: 600-612, (2004).
- [18]. Winderlich, Kristina, Clementine Dalelane and Andreas Walter. “Classification of synoptic circulation patterns with a two-stage clustering algorithm using the structural similarity index metric (SSIM).” (2022).
- [19]. Yan, Zhancai and Liu Yaqiu. “Image confusion algorithm using adaptive threshold SSIM based on Contourlet transformation.” Australian Journal of Mechanical Engineering 16: 23 – 29, (2018).
- [20]. Zheng, Xu, Tejo Chalasani, Koustav Ghosal, Sebastian Lutz and Aljosa Smolic. “STaDA: Style Transfer as Data Augmentation.” VISIGRAPP (2019).
- [21]. Zhou, Yubeibei, Jiulu Gong, Weijian Lu, Naiwei Gu, Kuiqi Chong and Zepeng Wang. “An object detection algorithm for military vehicles based on image style transfer and domain adversarial learning.” 2022 IEEE International Conference on Unmanned Systems (ICUS): 1059-1064, (2022).
- [22]. Zhu, Jun-Yan, Taesung Park, Phillip Isola and Alexei A. Efros. “Unpaired image-to-image translation using cycle-consistent adversarial networks.” 2017 IEEE International Conference on Computer Vision (ICCV): 2242-2251, (2017).
- [23]. Yolov8 is available at <https://github.com/ultralytics/ultralytics>.

TÓM TẮT

Tăng cường dữ liệu sử dụng phương pháp chuyển kiểu cho bài toán phát hiện đối tượng nguy trang quân sự

Phát hiện các vật thể nguy trang trong môi trường quân sự đặc biệt khó khăn do mục tiêu cố tình hoà trộn với môi trường xung quanh. Mặc dù các phương pháp phát hiện đối tượng sử dụng học sâu đã có sự tiến bộ lớn, nhưng sự hạn chế về số lượng và chất lượng của dữ liệu huấn luyện vẫn là một trở ngại đáng kể tới độ chính xác của mô hình phát hiện. Bài báo này đề xuất một phương pháp tăng cường dữ liệu mới kết hợp nhiều mô hình chuyển phong cách để chuyển đổi phong cách của ảnh huấn luyện thành nhiều phong cách tham chiếu giúp ảnh huấn luyện phong phú với nhiều kết cấu và mẫu môi trường khác nhau. Chúng tôi sử dụng độ tương đồng về cấu trúc (SSIM) là chỉ số đánh giá để chọn dữ liệu chuyển phong cách bảo toàn độ tương đồng về cấu trúc tốt nhất, cho phép các mô hình phát hiện học hiệu quả hơn các đặc điểm phân biệt của các vật thể nguy trang. Các thử nghiệm mở rộng với các phương pháp chuyển phong cách HierarchyFlow, CycleGAN và ArtFlow cùng với ngưỡng SSIM phù hợp cho thấy phương pháp tăng cường của chúng tôi cải thiện đáng kể độ chính xác của các thuật toán phát hiện hiện đại Yolov8. Phương pháp này có tiềm năng cải thiện khả năng phát hiện đối tượng trong các hoạt động quân sự, tăng độ tin cậy và độ chính xác của các hệ thống giám sát tự động.

Từ khoá: Phương pháp chuyển kiểu; Phát hiện đối tượng nguy trang quân sự; Tăng cường dữ liệu.